<u>SC435-Introduction to Complex Networks</u>

Course Title: Introduction to Complex Networks

Credit Structure(L-T-P-Cr): (3-0-0-3)

Course Code: **SC435**

Prerequisites (if any**)/desired skill set**: Much of the course is self-contained. However, a sound background in **Introductory Calculus**, **Probability and Statistics, Linear Algebra**, and **basic programming** is required. Some foundations in graph theory can be helpful but not required.

Level: Advanced Undergraduate

Course objective: Many systems spanning multiple disciplines are composed of many interacting components. Some examples of such systems are neurons in the brain, interaction of people, entities in markets, businesses, computers interacting over the internet, etc. Universality and emergence are two essential properties of such systems. Two directions of exploration, which are also intertwined, are analyzing large-scale data and constructing simple models. Complex networks provide a way to look at/develop understanding/explore/draw conclusions in such varied systems. Besides, it also forms the basis of many machine learning and deep learning methods. Therefore, These tools are essential for large-scale data analysis and mathematical and computational modeling professionals. This course aims to provide a rigorous foundation and practical implementations of the concepts learned.

Since many problems are interdisciplinary, the course has computer science, statistical physics, social sciences, and biology elements. This course aims to introduce students to many of these concepts in detail from the point of view of analytical and algorithmic, and computational methods. Various examples, from physical to social and biological systems, will be discussed. The course will cover three main topics of network science: (i) network structure and metrics, (ii) network models (iii) dynamical processes on networks.

The course includes a strong project component. The project aims to apply the class discussion to various online network databases. The fields can be any (social science, finance, biological systems, computer science, physics, etc.). Students are expected to work in small groups. The aim is to understand the system. The students use available data or, if it is a model study, generate relevant model-based data and/or implement, characterize the network, and identify what would be a relevant metric, algorithm, procedure, etc., to understand the network.

Evaluation Scheme: 60% exam (Mid semester + Final) + 40% project.

- **Exam:** One mid-semester (30%) and one final exam (30%).
- **Project:** The students would implement a relevant computational paper from the literature. The outcome of the project will be (i) documented and working code for the project (existing libraries can be used) (ii) report on the work carried out with proper citation to the references used to develop the understanding (iii) viva/presentation.
- **Attendance Policy:** Any student with less than **75%** of attendance in lectures will not be allowed to appear for the final exam.

- **Plagiarism and cheating policy:** Plagiarism and/or cheating will result in disqualification from the course.

Suggested textbook/references:

1. M. E. J.Newman, *Networks – An introduction.* Oxford Univ. Press, 2010.
2. A. L. Barabasi, *Network Science*, Cambridge, 2015.
3. F. Menczer, S. Fortunato, and C. A. Davis, *A First course in Network Science*, Cambridge University Press, 2020.

Reference Books

1. Alain Barrat, Marc Barthelemy, Alessandro Vespignany, Dynamical processes on complex networks, Cambridge University Press, 2012.
2. Eli Ben-Naim , Complex Networks (Lecture notes in Physics), Springer, 2004.
3. S. N. Dorogovtsev and J. F. F. Mendes, *Evolution of networks*, Oxford University Press, 2003.
4. Guido Calderelli, *Complex webs in nature and technology*, Oxford university Press, 2007.
5. Marc Newmann, Barabsi, Duncan Watts, *The structure and Dynamics of networks*, New Age International Pvt Ltd, 2010.

Course Outcomes:

The nature of the course is applied interdisciplinary. On successful completion of the course, students will learn

- to construct networks from data
- analyse the system of interest from the perspective of network science
- interdisciplinary research and education.

After succesfull completion of the course the student will have the ability to

- Apply the knowledge of mathematics and computer science to model complex problems in science (**P1,P2,P3,P12**).
- Carry out model based investigations to derive insight into the problem at hand **(P4).**
- Communicate effectively through vivas, course presentation and report writing **(P10)**

P1	**P2**	**P3**	**P4**	**P5**	**P6**	**P7**	**P8**	**P9**	**P10**	**P11**	**P12**
X	**X**		**X**	**X**				**X**			**X**

Lecture Schedule*

Sl. No.	**Description**	**No. of Lectures***
1	**Introduction to Complex Network: I**ntroduction to Networks, Empirical study of Real world networks (social, Technological, Biological, Information and neural networks). Concept of hubs.	**2-3**
2	**Basic Concepts in Graph Theory:** Network representation: adjacency matrix and edge lists, Types of Networks: Weighted, directed, bipartite, planar network, Graph Laplacian.	**5-6**
3	**Measures and metrics:** Degree, betweenness and closeness centrality, Eigenvector centrality, page rank and HITS. Katz centrality, Groups of	**5-7**

	vertices, Transitivity, Clustering coefficient, Reciprocity. Signed edges and structural balance, Similarity and mixing.	
4	**Community Structure:** Dividing network into groups, Modularity Maximization, Algorithms for Community Detection: Greedy algorithms, Spectral methods, Kernighan-Lin, Louvain Method, Hierarchical Clustering- Agglomerative (single-linkage, complete linkage and average linkage) and Divisive algorithms (edge betweenness method)	**4-5**
5	**Large scale structure of networks:** Components, shortest path and small world effect, Degree Distributions, Power laws and scale free networks.	**5-6**
6	**Models of Networks:** Random Networks: Properties of ER network. Configuration model: Model with arbitrary degree distribution, comparison with ER model. Barabasi Albert model: Preferential attachment model, mean field equation and master equation. Small world model (Watts-Strogatz).	**5-7**
7	**Dynamical Processes of networks:** Epidemic spread through basic compartmental models like SIS, SIR etc. on networks. Diffusion on networks, Resilience of networks.	**6-8**

* Total number of topics that will be covered will depend on the student participation and preparedness to understand new topics.